CS543 • WEB SERVICES

ASSIGNMENT 6 OF 12

Honest Responses: Status Codes, Errors & Validation

Group ID: 17

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Part A — Status Codes

AlgoViz API status mapping with Spring Boot code

A1. Short theory

HTTP status codes tell the frontend whether an API request succeeded or failed. The response status is the result: do not return 200 OK for a failed request with a body such as {"ok":false}. For a new visualization, use 201 Created and a Location header. Use 4xx for caller/request problems and 5xx for server-side failures.

A2. Endpoint status map

These are proposed paths based on the AlgoViz project description. Rename paths to match the actual controller routes in your code.

Endpoint	Success	Common failures
GET /api/algorithms	200 OK	401/403 if protected; 500 unexpected error.
POST /api/visualizations	201 Created	400 malformed JSON; 401 unauthenticated; 409 idempotency conflict; 422 invalid algorithm/input; 429 rate limit; 500/503 server fault.
GET /api/visualizations/{id}	200 OK	401 unauthenticated; 404 unknown/inaccessible run; 500 server fault.
GET /api/history	200 OK	401 unauthenticated; 500 server fault.
DELETE /api/history/{id}	204 No Content	401 unauthenticated; 404 missing/inaccessible record; 500 server fault.

A3. Spring Boot example: create visualization

```
@PostMapping("/api/visualizations")
public ResponseEntity<VisualizationResponse> create(
@Valid @RequestBody VisualizationRequest request) {
VisualizationResponse result = visualizationService.create(request);
URI location = URI.create("/api/visualizations/" + result.id());
return ResponseEntity.created(location).body(result);
}
```

A4. Example DTOs

```
public record VisualizationRequest(
@NotBlank String algorithm,
@NotEmpty List<Integer> input) {}

public record VisualizationResponse(
String id,
String algorithm,
List<Integer> input,
String status) {}
```

A5. Retry guidance

```
// For 429/503 responses, include Retry-After.
// Example: return ResponseEntity.status(503)
// .header(HttpHeaders.RETRY_AFTER, "5")
// .body(problemBody);
```

A6. Retry header example


```
// For 429/503 responses, include Retry-After.
// Example: return ResponseEntity.status(503)
// .header(HttpHeaders.RETRY_AFTER, "5")
// .body(problemBody);
```


Notes / implementation caution

A successful POST returns 201 and Location. If your backend queues work, use a documented 202 Accepted contract instead. Do not copy both approaches into the same endpoint without deciding which one your service uses.